Project Title

Tourist Risk Analytics & Incident Live-Response (T.R.A.I.L.)

1. Introduction

1.1. Purpose

The purpose of this document is to specify the functional and non-functional requirements of T.R.A.I.L. (Tourist Risk Analytics & Incident Live-Response), an AI-assisted tourist safety monitoring and emergency response system. This SRS serves as the primary reference for system design, development, testing, and academic evaluation.

1.2. Scope

T.R.A.I.L. is an integrated mobile and web-based system designed to enhance tourist safety in the North-Eastern region of India. The system provides secure digital tourist identification, real-time location monitoring using geo-fencing, AI-based risk and anomaly detection, and emergency alert mechanisms connected to a centralized authority dashboard.

1.3. Definitions, Acronyms, Abbreviations

Term	Meaning
T.R.A.I.L.	Tourist Risk Analytics & Incident Live-Response
AI	Artificial Intelligence
ML	Machine Learning
GPS	Global Positioning System
Geo-Fence	Virtual geographic boundary
SOS	Emergency Distress Signal
Anomaly	Abnormal movement or behaviour

1.4. Intended Audience

- Project evaluator / faculty guide
- Software engineering students
- Developers implementing the system
- Tourism and safety authorities

2. Overall Description

2.1. Product Perspective

T.R.A.I.L. is a standalone distributed system consisting of:

- A mobile application for tourists
- A web-based monitoring dashboard for authorities
- Backend services handling AI analysis, geo-processing, and data storage

The system is designed for demonstration and academic evaluation without real-world enforcement integration.

2.2. Product Functions (High-Level)

- Digital tourist registration and ID generation
- Real-time location tracking
- Geo-fencing and risk zone alerts
- AI-based anomaly and risk detection
- SOS-based emergency reporting
- Authority-side monitoring and analytics

2.3. User Classes

User	Description
Tourist	Registers, travels, receives alerts, sends SOS
Admin / Authority	Monitors tourists, incidents, and analytics

2.4. Operating Environment

- Mobile platform: Android smartphone
- Web platform: Modern web browsers
- Backend hosted on cloud infrastructure
- Centralized database with geo-location support

2.5. Design and Implementation Constraints

- Academic prototype development
- Limited development duration
- Pre-trained AI/ML models
- No real-time police or rescue system integration

2.6. Assumptions and Dependencies

- Tourists carry GPS-enabled smartphones
- Continuous internet connectivity is available
- Mapping and notification APIs are accessible
- AI outputs are probabilistic in nature

3. System Features & Functional Requirements

3.1. Digital Tourist Identity System

Description:
The system authenticates tourists and assigns a secure digital identity for tracking and safety monitoring.

Functional Requirements:

- FR1: The system shall allow tourists to register using basic personal details.
- FR2: The system shall generate a unique Digital Tourist ID for each registered user.

3.2. Smart Tourist Safety Mobile Application

Description:
The mobile app enables tourists to view alerts, share location, and request emergency assistance.

Functional Requirements:

- FR3: The system shall continuously capture tourist location using GPS.
- FR4: The system shall allow tourists to trigger an SOS alert manually.

3.3. Geo-Fencing & Risk Zone Monitoring

Description:
The system monitors tourist movement across predefined safe and high-risk zones.

Functional Requirements:

- FR5: The system shall define geo-fenced zones for monitored areas.
- FR6: The system shall generate alerts when tourists enter restricted or risky zones.

3.4. AI-Based Risk & Anomaly Detection

Description:
AI models analyse tourist movement patterns to detect unusual or potentially dangerous behaviour.

Functional Requirements:

- FR7: The system shall analyse location patterns using AI models.
- FR8: The system shall detect anomalies such as route deviation or prolonged inactivity.
- FR9: The system shall assign a risk score to each tourist session.

3.5. Emergency Alert & Incident Response

Description:
High-risk situations trigger alerts visible to authorities in real time.

Functional Requirements:

- FR10: The system shall generate alerts for high-risk or SOS events.
- FR11: The system shall notify authorities through the monitoring dashboard.

3.6. Authority Monitoring Dashboard

Description:
Authorities can monitor tourist locations, risks, and incidents using a centralized interface.

Functional Requirements:

- FR12: The system shall display live tourist locations on a map.
- FR13: The system shall provide real-time alerts and notifications.
- FR14: The system shall display historical movement and incident data.

4. External Interface Requirements

4.1. User Interfaces

- Mobile application interface for tourists
- Web-based dashboard for authorities
- Map-based visualization of locations and alerts

4.2. Software Interfaces

- Map and GPS APIs
- AI/ML inference libraries
- Notification services
- Database management system

5. Non-Functional Requirements

5.1. Performance

- NFR1: Location updates shall occur in near real time.
- NFR2: SOS alerts shall be processed immediately.

5.2. Reliability

- NFR3: The system shall remain operational during continuous monitoring sessions.

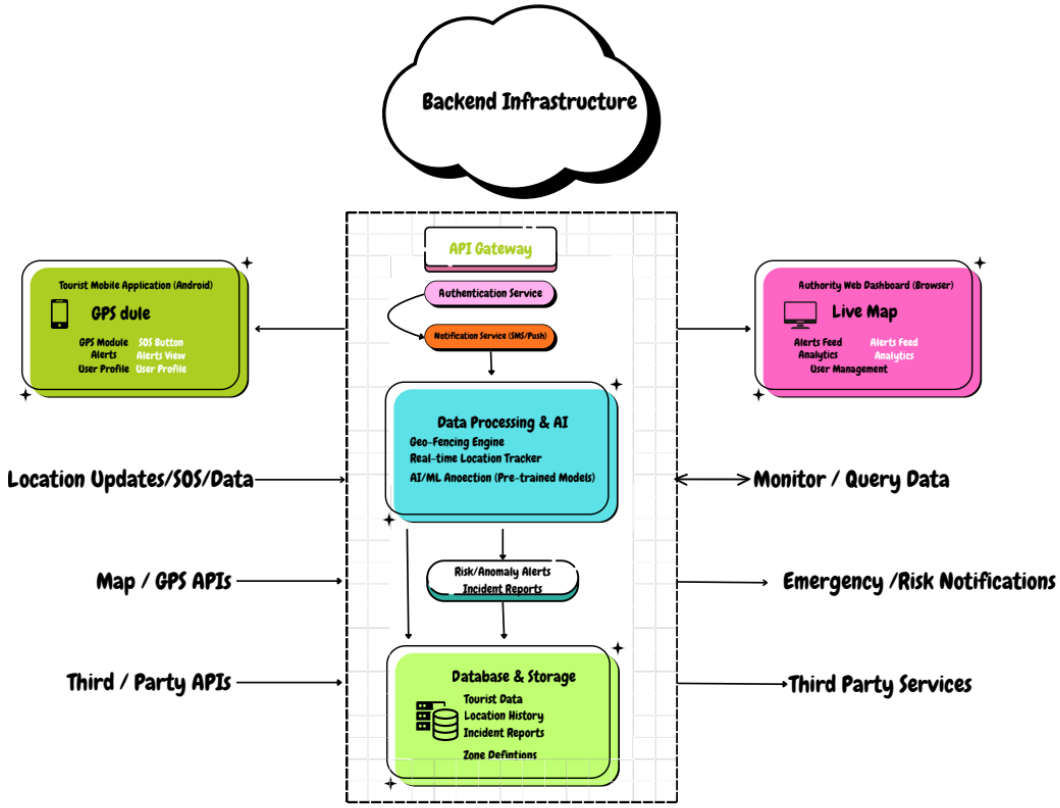
5.3. Security

- NFR4: Only authenticated users shall access the system.
- NFR5: Sensitive user data shall be securely stored.

5.4. Scalability

- NFR6: The system shall support a growing number of tourists.
- NFR7: AI processing services shall scale independently.

5.5. Usability

- NFR8: The mobile app shall be simple and user-friendly.
- NFR9: Emergency actions shall be executable within seconds.

6. Architecture Diagram